

# Zhiheng Zhang

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github.com/zhiheng-zhang-Mera

Melbourne, Australia

## Research Profile

Computer Science research engineer focused on pervasive intelligence, multi-device agent systems and reliable edge execution.

## Education

### University of Melbourne

*Master's degree in Computer Science*

Melbourne, Australia

*Expected 2026*

### University of British Columbia

*Bachelor of Science in Computer Science*

Canada

*Completed Summer 2024*

## Selected Research and Engineering Projects

### Utopia / Digital City — *Multi-device personal-computing and research platform*

2026–present

github.com/zhiheng-zhang-Mera/Utopia

- Built Android and Web control surfaces around a platform-neutral City protocol with authenticated device/control paths, explicit confirmation and truthful degraded-state handling.
- Current product evidence includes accepted V0.2/V0.3 boundaries and an accepted three-end mesh across two real Windows workers plus an Android control surface with strict target-device routing and cross-host review.
- The research-strengthening programme has accepted experiment registry, trace/provenance, repeated-scenario, fault-injection, and replay/ablation stages; artifact export remained under independent review at the 2026-10-06 snapshot.
- Next research direction: a **Personal Compute Fabric** for heterogeneous-device telemetry, placement/offloading, recovery and explainable cross-device execution; this is a planned extension, not claimed complete.

### Codex Boss — *Agent orchestration and evidence-based acceptance*

2026–present

github.com/zhiheng-zhang-Mera/Codex-Boss

- Developed a multi-agent workflow platform for planning, execution, critique and adjudication with durable state and evidence-aware acceptance.
- Treats unresolved failure cases and incomplete acceptance conditions as explicit evidence rather than reporting them as solved.
- Supports research on trustworthy agent orchestration, completion claims, auditability and human oversight.

### DS-Hns — *Long-horizon autonomous software-work runtime*

2026–present

github.com/zhiheng-zhang-Mera/DS-Hns

- Building long-horizon repository execution with durable task state, restart/recovery, hardware-aware concurrency and explicit terminal-state evidence.
- Uses failure isolation and recovery-oriented execution to study when a software agent has genuinely completed work rather than merely survived as a process.
- Supports research on coding-agent reliability, autonomous maintenance, AIOps and empirical software-engineering evaluation.

## Research Interests

Pervasive intelligence; Physical AI systems; edge/distributed AI; multi-device execution; reliable agent infrastructure.

## Technical and Research Strengths

- End-to-end research engineering: system design, implementation, testing, debugging, controlled experiments and evidence-driven iteration.
- Multi-device and long-running software systems with explicit lifecycle, recovery, routing, observability and acceptance boundaries.
- Git/GitHub-based reproducible development, regression testing, cross-host review and structured preservation of failure evidence.

## Current Research Direction

I want to study the systems layer between agent reasoning and physical/device action: routing, recovery, user control, provenance and resource-aware execution across heterogeneous endpoints.